



Modeling unknowns: A vision for uncertainty-aware machine learning in healthcare

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ABSTRACT

The integration of machine learning (ML) into healthcare is accelerating, driven by the proliferation of biomedical data and the promise of data-driven clinical support. A key challenge in this context is managing the pervasive uncertainty inherent in medical reasoning and decision-making. Despite its recognized importance, uncertainty is often underrepresented in the design and evaluation of clinical AI systems.

Here we report an editorial overview of a special issue dedicated to uncertainty modeling in medical AI, which gathers theoretical, methodological, and practical contributions addressing this critical gap. Across these works, authors reveal that fewer than 4% of studies address uncertainty explicitly, and propose alternative design principles—such as optimizing for clinical net benefit or embedding explainability with confidence estimates. Notable contributions include the RelAI system for real-time prediction reliability, empirical findings on how uncertainty communication shapes clinical interpretation, and benchmarks for out-of-distribution detection in tabular data. Furthermore, this issue highlights the use of causal reasoning and anomaly detection to enhance system robustness and accountability.

Together, these studies argue that representing, communicating, and operationalizing uncertainty are essential not only for clinical safety but also for building trust in AI-driven care. This special issue thus repositions uncertainty from a limitation to a foundational asset in the responsible deployment of ML in healthcare.

Introduction

In recent years, the clinical domain has been transformed by an unprecedented growth in the availability of biomedical and health-related data [1]. This development has spurred the adoption of machine learning (ML) methods to assist clinical decision-making, from diagnosis to prognosis and treatment recommendation [2]. Despite promising results in experimental settings, the widespread deployment of ML systems in real-world clinical environments remains limited. A key reason lies in the inability of many such systems to explicitly model, quantify, and communicate uncertainty—an intrinsic feature of medical data and medical reasoning [3–5].

Uncertainty modeling refers to the systematic representation, quantification, and propagation of incomplete, ambiguous, imprecise, or incorrect information within a computational or formal framework to support reasoning and decision-making [6]. In the context of medical informatics, it denotes the development and application of mathematical or algorithmic methods to explicitly capture and communicate the degree of confidence—or lack thereof—associated with diagnostic hypotheses, prognostic estimations, or therapeutic recommendations [7,8], as well as to model and manage complex, potentially unreliable or otherwise low-quality medical data in the model building process [9,7]. Uncertainty modeling thus encompasses both *epistemic uncertainty* (due to model limitations, incomplete knowledge, or lack of data) and *aleatoric uncertainty* (due to inherent variability or noise in observations) [10],

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and it aims to ensure that these uncertainties are quantified, managed and made explicit in a form interpretable by either human users or downstream decision algorithms. A robust definition, grounded in the literature on artificial intelligence and decision science, considers uncertainty modeling not merely as a computational accessory, but as a foundational component of any system designed to interact with real-world clinical variability and incomplete knowledge [11,12]. As such, it is increasingly recognized as essential for the design of trustworthy, safe, and clinically useful AI systems in healthcare [5,12].

This special issue on *Machine Learning and Uncertainty Modeling in Healthcare* gathers a set of contributions that approach the challenge of uncertainty not as a residual error to be minimized, but as a structural and epistemic dimension to be acknowledged, modeled, and rendered actionable. The included articles span foundational, methodological, and applied perspectives, illustrating how uncertainty-aware AI can enhance safety, foster human-AI collaboration, and support clinical accountability. Together, they demonstrate that rigorous uncertainty modeling is essential not only for predictive accuracy, but for the ethical and practical integration of ML technologies into real-world medical decision-making [3,5,12].

Before delving into the details of the works collected in this special issue, we provide a concise primer on the subject to help readers better appreciate their originality and the value they add to the existing literature.

Frameworks for uncertainty modeling

A variety of modeling frameworks have been developed to handle uncertainty in clinical decision-making. Each framework provides a formalism for representing and reasoning with uncertain information, whether it comes from data noise, missing knowledge, or conflicting evidence. Below we outline key approaches and their roles in medical informatics:

Bayesian inference and probabilistic models. Bayesian methods use probability theory as a language for uncertainty: unknown quantities (such as a patient's true diagnosis or the effect of a treatment) are treated as random variables with probability distributions that represent one's degree of belief [13]. By incorporating prior knowledge and new evidence, Bayesian inference updates these probabilities to form posterior beliefs. This approach is naturally suited to clinical reasoning under uncertainty [14]: for example, a Bayesian diagnostic model can start with prior probabilities of diseases and update them as lab tests and symptoms are observed. Probabilistic graphical models (which include Bayesian networks and Markov models [15], as well as Bayesian Neural Networks [16]) allow complex dependencies to be modeled and uncertainty to be propagated through those dependencies in a mathematically rigorous way [17]. While exact inference in large models can be computationally intensive, but various approximation techniques (e.g., Monte Carlo inference [18]) exist to make them tractable in practice.

Conformal prediction. In modern machine learning, conformal prediction is a statistical framework for uncertainty quantification, originally introduced by Vovk, Shafer, and Gammerman [19], drawing on ideas from online learning, statistical interval analysis, and algorithmic information theory. In contrast to probabilistic and Bayesian approaches, conformal prediction is distribution-free, meaning that it provides theoretical guarantees that hold regardless of the underlying data-generating process, provided that the data are independent and identically distributed, or more generally, exchangeable [20]. Conformal prediction expresses uncertainty through so-called *prediction sets* [21], which include all candidate outcomes (e.g., possible diseases in a diagnostic task, or potential outcomes in a prognostic task) that cannot be ruled out for the case at hand, like in a sort of clinical differential diagnosis [22]. These sets can be interpreted similarly to confidence intervals, in that they are associated with a confidence level quantifying the probability

that the prediction set contains the true value of the quantity of interest [23]. Prediction sets are constructed using a non-conformity measure [24], a function that quantifies how well a new instance *conforms to* (i.e., is similar to) previously observed instances. Unlike approaches that require ad-hoc model modifications, conformal prediction is a post-hoc, model-agnostic uncertainty quantification method [25], meaning it can be applied to augment any existing machine learning model with uncertainty estimates by defining an appropriate non-conformity measure.

Ensemble learning. In machine learning, one practical way to assess model uncertainty is to use an ensemble of models [26]. The idea is to train multiple models (e.g. multiple decision trees, or multiple neural networks with different initializations) and then aggregate their predictions: this provides an approximate, but practically effective, Bayesian inference method [27]. The variance among the ensemble's predictions can be used as a measure of uncertainty [28]: if all models confidently agree on an outcome, the prediction is likely reliable; if they disagree widely, it indicates high model uncertainty. Ensemble methods are popular in medical AI because they often improve accuracy while also providing an implicit uncertainty estimate (via diversity of outputs). For example, an ensemble of predictive models could be used for risk stratification: if one patient's risk of a complication varies from 5% to 30% across different models, whereas another patient's risk is consistently 15%, the first case has higher predictive uncertainty. Techniques like bootstrapping (training each model on a slightly different sampled dataset) further enhance the ensemble's ability to quantify uncertainty. In practice, clinicians can be shown not just a single risk score but a range or distribution coming from an ensemble, conveying the confidence in the prediction [29]. Ensemble learning is relatively straightforward to implement and often highly effective; however, it increases computational cost linearly with the number of models and can be harder to interpret (since many models contribute to the result).

Fuzzy logic. Clinical medicine often involves vagueness – many findings are not simply “present or absent” but rather a matter of degree [30] (e.g. a symptom might be mild, moderate, or severe). Fuzzy sets and fuzzy logic [31] provide a framework to handle such imprecision by allowing partial membership in sets and reasoning with degrees of truth. In fuzzy logic-based systems, clinical rules can be formulated with linguistic terms (like “if blood pressure is high and patient is young, then risk is moderately elevated”), and the uncertainty in those terms is mathematically managed by membership functions [32]. This approach is adept at handling uncertainty, imprecision, and vagueness in clinical data. A fuzzy logic system does not output a hard yes/no; instead it produces a nuanced output (often a fuzzy set or a confidence level) that reflects the uncertainty in the conclusion. In practice, fuzzy clinical decision support systems have been developed for tasks such as diagnosing diseases or evaluating patient risk [33]. By encoding medical knowledge in a rule-based manner and allowing partial truths, fuzzy logic systems can emulate a clinician's reasoning under uncertainty, especially in domains where exact probabilistic models are hard to specify.

Imprecise probabilities. Imprecise probabilities are a general collection of uncertainty modeling frameworks that generalize probability theory, by allowing to quantify uncertainty not just in terms of a single probability distribution but, more generally, of sets of probabilities [34]. Such frameworks stem from a discussion of the limitations of traditional probability theory, including the inability to properly model ambiguity and lack of knowledge. Among imprecise probabilities, two popular approaches are *Credal Set Theory* and *Dempster–Shafer Theory of Evidence*. Credal set theory (CST) [35], also known as lower previsions' theory [36], refers to a generalization of probability theory in which uncertainty is represented in terms of convex sets of probabilities (so-called credal sets), or, equivalently, as upper and lower probabilities. A key advantage of credal sets is that they enable explicit modeling of epistemic uncertainty by avoiding the need to commit all belief to a precise

probability distribution (e.g., evidence about some biomarker of interest could of insufficient quality to precisely identify a unique statistical model). CST then provides a generalization of Bayesian inference methods, including conditioning and probabilistic models [37]. In medical AI, credal sets have been used both as a way to model uncertainty in data, specifically to represent biological and analytical variability in laboratory measures [38], as well as to implement more reliable models that can abstain from providing a prediction in cases of uncertainty [39]. Dempster–Shafer theory (DST) [40], also known as the theory of belief functions and evidence theory [41], is a framework for reasoning with uncertainty that generalizes Bayesian probability. DST allows one to represent degrees of belief over subsets of possibilities, providing a way to model both uncertainty and ignorance [42]. One key feature is that it permits assigning belief to composite hypotheses (e.g. a clinician might not be sure which of two diseases is present, but is confident it's one of them). Evidence from different sources can then be combined using Dempster's rule of combination, which accounts for the support each piece of evidence gives to various hypotheses. In medical diagnosis, DST can be used to aggregate information from multiple tests or observers: for instance, if two diagnostic tests each have uncertain sensitivity and specificity, DST can combine their results to calculate an overall belief in each possible diagnosis while explicitly representing the remaining uncertainty (the belief mass that is uncommitted). This is advantageous in cases where information is incomplete – DST will not force a false certainty but instead acknowledges the “unknown” portion of belief. Overall, imprecise probabilities provide a powerful alternative to traditional probability for managing uncertainty in clinical domains, though their application in complex settings can be difficult from a computational point of view [43–45].

PAC learning. In machine learning, particularly within theoretical computer science, determining whether and how a concept can be learned is a foundational question. The Probably Approximately Correct (PAC) learning framework, introduced by Leslie Valiant [46], provides a formal basis for analyzing the feasibility of learning from data under uncertainty. In the PAC model, a learner seeks a hypothesis that, with high probability, performs nearly as well as the best possible function in a given class, based on a finite set of training examples drawn from an unknown distribution. A central aspect of PAC learning is its reliance on probabilistic guarantees: a concept is said to be PAC-learnable if, for any small values of ϵ (accuracy) and δ (confidence), there exists a learning algorithm that, with probability at least $1 - \delta$, returns a hypothesis whose error is at most ϵ . The number of training examples required must be polynomial in $1/\epsilon$, $1/\delta$, and the complexity of the concept class. This framework abstracts away from the specifics of individual learning algorithms and instead addresses the fundamental feasibility of learning. PAC learning prioritizes generalization over mere data fitting. It highlights the trade-off between hypothesis complexity and sample size [47], and forms the basis for theoretical tools such as the Vapnik-Chervonenkis dimension [48], which measures the capacity of a hypothesis space. Although real-world data often violate PAC assumptions (e.g., due to noise or non-stationarity), the framework remains central to the theoretical foundations of algorithm design, and several generalizations have been proposed to extend its applicability to more complex settings [49–51].

Applications in clinical settings

In real-world healthcare settings, uncertainty modeling is being applied to enhance decision support across a range of tasks. Below are several key areas in which uncertainty-aware models improve clinical decision-making, along with illustrative examples:

Diagnostic support systems. Clinical diagnostic tools increasingly provide probability estimates or confidence scores for different diagnoses

instead of a single definitive output. This helps clinicians weigh the differential diagnosis alongside the model's certainty. For instance, an AI system for skin lesion analysis might output a probability distribution over possible conditions (benign, melanoma, etc.) with error bars, rather than a binary verdict. In medical imaging, uncertainty maps can be used to identify regions where an algorithm is less certain – “the uncertainty in the model's predictions can be used to identify areas of concern or to provide additional information to the clinician” in making a diagnosis. As a concrete example, if a chest X-ray analysis model is unsure whether an opacity is pneumonia or an artifact, it can flag that uncertainty, prompting the radiologist to gather more information (such as another imaging view or clinical correlation). Overall, by presenting probabilities, confidence intervals, or other expressions of uncertainty such as “abstaining” when unsure, diagnostic support systems become safer and more useful partners to human clinicians. Models that know when they don't know can suggest second opinions or additional tests, mirroring the human physician's approach to ambiguity. This leads to more robust and transparent diagnostic decision-making.

Risk stratification and prognostic modeling. Predictive models are widely used to stratify patients by risk (for example, risk of developing a complication, readmission risk, or disease progression risk) and to forecast outcomes (prognosis). Incorporating uncertainty into these predictions is crucial for proper interpretation. Clinicians are more likely to trust and act on a risk score if they understand its confidence bounds. For instance, imagine a model that predicts a 5-year mortality risk for a cardiac patient. Rather than just giving a point estimate like “20% risk,” an uncertainty-aware model might report that the risk is 20% (with a 95% confidence interval of 10–30%). This communicates the model's confidence and can alter decision thresholds (a wide interval might suggest waiting or gathering more data, whereas a narrow interval gives more actionable information). A practical example arises in the setting of heart disease prediction: assume an ensemble of models is used to generate a distribution of risk scores for each patient. Two patients with similar mean risk (say 15%) may be found to have very different uncertainties – one with a tight risk distribution, the other a very broad one. The patient with the broad, “dispersed” risk distribution could be “flagged as needing more information before making a clinical decision”. Such an approach ensures that patients whose risk is highly uncertain are identified for further assessment (additional tests, closer follow-up, or expert review), thus reducing the chance of unforeseen adverse outcomes. In prognostic modeling (e.g. predicting cancer survival or disease progression curves), uncertainty estimates translate into prediction intervals or confidence bands around survival curves. Clinicians can use these to discuss best-case and worst-case scenarios with patients and to plan treatments that are robust against uncertainty. Uncertainty modeling in risk and prognostic tools ultimately leads to better stratification and tailored care, as decisions (like starting a therapy or recommending surgery) can be adjusted based on not just the predicted risk but also the confidence in that prediction.

Therapeutic decision-making and treatment recommendations. Recommending an optimal treatment involves weighing not only expected benefits and risks but also the uncertainty in those estimates. Decision support systems that incorporate uncertainty can identify when a particular treatment recommendation is marginal or when patient-specific data is too sparse to be confident. For example, in personalized medicine, a model might suggest Treatment A has a slightly higher expected success rate than Treatment B for a given patient – but if the uncertainty (variance) around A's outcome is large, whereas B's outcome is more certain, a clinician might choose the more robust option B. This approach allows physicians to select treatments with lower variance when necessary, effectively favoring more predictable outcomes under high uncertainty. In critical care and chronic disease management, where treatment policies can be modeled as sequential decisions, algorithms have been developed to factor in model uncertainty and even

defer to human judgment when confidence is low. In practice, a treatment decision support tool might highlight that two treatments have statistically similar expected outcomes given the data, signaling that uncertainty is high and patient preferences or further tests might be needed to decide. Or, conversely, it might show one treatment is superior with high confidence. By quantifying uncertainty in treatment effects, clinicians and patients can engage in informed shared decision-making, understanding the range of possible outcomes. This is especially pertinent in scenarios like novel therapies or rare diseases, where data is limited (epistemic uncertainty high) and a cautious interpretation of model recommendations is warranted.

Challenges and limitations

While uncertainty modeling offers clear benefits for medical decision-making, its practical implementation comes with challenges:

Computational complexity. Many uncertainty modeling techniques are computationally intensive. Bayesian inference in complex models or deep Bayesian neural networks often requires approximation methods (e.g. Monte Carlo sampling) that can be slow. Similarly, ensemble methods demand training and running multiple models instead of one, which can strain computational resources especially with large datasets or real-time requirements. For example, a full Bayesian neural network for an ICU prognostic model might be theoretically ideal to capture uncertainty in model weights, but performing posterior inference over millions of parameters is infeasible in real-time. Researchers have addressed this by using approximations. Nonetheless, even these solutions can introduce latency. In clinical settings, there may be a trade-off between the depth of uncertainty analysis and the speed of decisions – waiting an extra 30 minutes for an elaborate uncertainty quantification might not be acceptable in an emergency. Therefore, optimizing algorithms for efficiency, and leveraging hardware acceleration, are active areas of development to ensure uncertainty estimates can be obtained in a timely manner.

Interpretability and communication. Providing uncertainty information is only useful if clinicians (and patients) can understand and act on it. This raises challenges in how to present uncertainty. Healthcare professionals are accustomed to certain formats (e.g. confidence intervals, risk categories), but not all are statistically trained to interpret, say, a predictive probability density function or a mass function. Models that produce too complex an uncertainty output risk overwhelming or confusing end-users. Moreover, many current AI models in literature “do not report sample-level uncertainty metrics such as calibration” and often have no mechanism to indicate when they are unsure. This reflects a gap in both model development and user-interface design. Ensuring that uncertainty estimates are well-calibrated (so that, for example, a predicted 20% risk truly corresponds to 1 in 5 patients having the event) is critical; poorly calibrated models can mislead clinicians into false confidence or undue alarm. Another issue is deciding thresholds for action: if a model says it is “not confident,” should the system automatically defer to a human or ask for more tests? These workflow questions require careful thought and are the subject of ongoing research into communicating uncertainty. While visual aids (like error bars, prediction intervals on plots, or color-coded uncertainty heatmaps) and allowing the model to abstain when uncertainty is high can engender greater trust, still, integrating such features into electronic health records or decision support interfaces in a user-friendly way is not trivial. The medical community will need training and guidelines on interpreting AI uncertainty – analogous to how we educate clinicians to interpret diagnostic test sensitivity, specificity, and confidence intervals in research.

Data quality and availability. The old adage “garbage in, garbage out” applies strongly to uncertainty models. If the data used to train or feed the model is of poor quality, the uncertainty estimates themselves may be unreliable. Missing data, measurement errors, outliers, and biases

in datasets all contribute to uncertainty in outputs. While uncertainty modeling techniques can sometimes flag these issues (for example, a large epistemic uncertainty might signal insufficient data in a certain patient subgroup), they do not magically overcome fundamental data deficiencies. In healthcare, data are often messy – patients might have incomplete histories, or measurements might come from different devices with varying accuracy. Furthermore, distributional shift between training data and real-world data can greatly increase uncertainty. As an example, a predictive model for sepsis trained in one hospital may encounter a very different patient population elsewhere; if not detected, this shift can lead to overconfident but wrong predictions. It’s important to note that uncertainty estimates themselves can be uncertain if the model for uncertainty is mis-specified. For example, a Bayesian model’s credibility interval is only valid if the model’s assumptions hold; if there are unmodeled effects or unknown unknowns, even the uncertainty quantification can be misleading. There is also a lack of ground truth for uncertainty – we rarely know the true probability of an outcome for an individual patient, which makes validating and calibrating uncertainty estimates challenging. Researchers often resort to using large test sets to see if, say, 95% of patients indeed have outcomes within the model’s 95% prediction interval. Ensuring good data practices (cleaning, outlier detection, bias mitigation) is a prerequisite for meaningful uncertainty modeling. Moreover, obtaining sufficient high-quality data (for example, enough examples of rare events to reduce epistemic uncertainty) may require multi-center collaborations or novel trial designs. In short, data quality issues and dataset shifts remain major hurdles in reliable uncertainty quantification.

Integration into clinical workflow. Finally, even if we have a well-characterized uncertainty from a model, using it in practice requires workflow integration and acceptance by clinicians. This is as much a socio-technical challenge as a technical one. Clinicians may have varying tolerance for uncertainty – some might appreciate a model saying “I’m not sure,” while others might find it frustrating or prefer a second opinion from a colleague over a statistical measure. Clinical decision support tools must be carefully validated and likely regulatory approved, especially if they advise actions (particularly true if a model will abstain or suggest further tests based on uncertainty). The human factors of how uncertainty advice is delivered (does it appear as an alert? as a confidence percentage? as a narrative explanation?) can influence whether it is heeded or ignored. Without careful design, there is a risk that additional uncertainty information could contribute to alert fatigue or be misinterpreted. Early evidence shows that when done properly, conveying uncertainty can improve trust – doctors feel more comfortable knowing the model’s limits and seeing that it aligns with their own cautious approach. But achieving this requires iterative development with end-user feedback, education, and possibly cultural shifts in how we view clinical AI (from omniscient oracles to fallible assistants).

Contributions of this special issue

This special issue opens with the critical review by Cabitza and Marconi [52], which synthesizes the challenges and limitations surrounding robustness and uncertainty in medical AI. Their bibliometric and conceptual analysis reveals a striking gap between the recognized importance of uncertainty quantification and its actual implementation: fewer than 4% of studies in leading medical AI journals explicitly address external validation or model uncertainty. Beyond merely reporting this shortfall, the authors advocate for institutional reform—urging journals, regulators, and developers to enforce higher methodological standards and to foster a culture in which uncertainty is not hidden but foregrounded. Their call to “show and tell” uncertainty aligns closely with the editorial rationale of this special issue: that explicitly modeling and communicating uncertainty is not a sign of weakness, but a prerequisite for trust, safety, and meaningful integration into clinical care.

Building on this normative foundation, Vickers et al. [53] propose a paradigm shift in how clinical machine learning models are developed and evaluated. They argue that model training should be aligned with clinical objectives from the outset—specifically, by optimizing for net benefit, a decision-theoretic metric that directly reflects the trade-offs between true and false positives in clinical decision-making. Rather than treating clinical utility as a post hoc evaluation criterion, their framework integrates it into the core optimization process. This proposition challenges the assumption that statistical metrics such as AUC or accuracy naturally correlate with real-world utility, and reframes the machine learning lifecycle around the value it delivers to end-users. The conceptual implications are significant: clinical AI is positioned not as an abstract mathematical exercise but as a normative, outcome-oriented discipline.

While Vickers et al. provide a theoretical reorientation, Salvi and colleagues [54] address a persistent gap in the literature by bridging the domains of explainability and uncertainty quantification. Their position paper argues convincingly that these two goals should not be treated in isolation. Instead, they propose that uncertainty estimation can serve to qualify and contextualize explanations—helping to assess not only what a model predicts and why, but also how confidently it does so. Their integrative approach mitigates the risk of overconfidence in apparently intuitive explanations and underscores that interpretability without uncertainty awareness may be misleading. By illustrating this point through practical examples in deep learning, they elevate uncertainty quantification from a backend diagnostic tool to a first-class citizen in the interpretability discourse.

This conceptual synthesis is operationalized in the work of Peracchio et al. [55], who introduce *RelAI*, a practical tool for evaluating the reliability of individual predictions at the point of care. *RelAI* combines autoencoder-based outlier detection with local performance estimation to flag potentially unreliable outputs in real-time clinical deployment. Its model-agnostic design and seamless compatibility with standard MLops pipelines make it especially promising for integration into existing workflows. Validated on a real-world multiple sclerosis dataset, the system identifies contextual conditions under which predictions may fail silently—highlighting the importance of prediction-level uncertainty. *RelAI* exemplifies how theoretical principles of uncertainty modeling can be translated into actionable functionalities that directly support clinical judgment and enhance trust in AI recommendations.

The need to identify and screen potentially unreliable predictions is also evident in the work of Khanizadeh et al. [56], who develop a dual-strategy anomaly detection framework tailored to healthcare time series data. Their method distinguishes between collective anomalies—patterns that deviate across institutions—and individual anomalies confined to specific care centers. By employing unsupervised learning techniques and interactive visual analytics, their system operates without requiring labeled data, making it suitable for data-sparse or highly heterogeneous environments. What makes their contribution particularly relevant is its dual epistemological stance: it recognizes the limitations of predictive models and simultaneously equips practitioners with tools to detect failures and surprises. This underscores the idea that trustworthy AI in healthcare must be not only accurate but also vigilant, adaptive, and context-aware.

Continuing the focus on detecting anomalous and unreliable data, Azizmalayeri et al. [57] address the challenge of identifying out-of-distribution (OOD) inputs in medical tabular data—a critical yet underexplored source of uncertainty in clinical AI. They present a reproducible benchmarking framework and evaluate a diverse set of OOD detection methods across four large public health datasets. Their results are revealing: while some methods perform adequately under clear distributional shifts, most fail to detect more subtle shifts related to demographic attributes such as age or ethnicity. These findings have significant implications for fairness, generalizability, and safety in model deployment. The authors argue persuasively that robust OOD detection must be sensitive not only to statistical divergence but also to clinical

relevance. In doing so, they reaffirm a central theme of this special issue: that uncertainty modeling must account for both technical and contextual complexity.

Alptekin et al. [58] extend the conversation into the domain of human factors, studying how different ways of presenting uncertainty influence clinical interpretation. Using a Bayesian network model for trauma-induced coagulopathy, they test how probabilistic predictions and performance metrics—whether verbal, graphical, or numerical—affect clinicians' risk perception and decision-making. Their findings are sobering: even subtle changes in the presentation format can lead to significantly different interpretations, especially in borderline or ambiguous cases. The perceived reliability of the model is shown to depend not only on its internal performance but also on how that performance is externally communicated. Their study supports the broader thesis that uncertainty is not solely a technical attribute of algorithms but also a communicative property of user interfaces—one that shapes trust, comprehension, and ultimately clinical action.

Adopting a similar computational framework, Kyrimi et al. [59] explore a complementary dimension of uncertainty through the lens of counterfactual reasoning. Leveraging causal Bayesian networks, they introduce a method for incorporating counterfactual analysis into healthcare governance practices, particularly within morbidity and mortality review processes. Their framework allows clinicians to evaluate alternative clinical trajectories—what might have occurred under different interventions—and thereby to institutionalize a form of structured reflection. Here, uncertainty is not merely probabilistic but narrative and epistemic: it enables clinicians to engage with the limits of their knowledge and to extract actionable insights from hypothetical scenarios. This work demonstrates that uncertainty modeling can support not only real-time decision-making but also retrospective evaluation and quality assurance.

Finally, Sblendorio et al. [60] address uncertainty from a safety and regulatory perspective by evaluating the feasibility of using large language models (LLMs) in clinical oncology nursing. Their evaluation framework combines multi-dimensional performance metrics with domain expert reviews, assessing models across a range of criteria including terminological consistency, prompt reliability, data safety, and epistemic soundness. Their results are sobering: while a few state-of-the-art models qualify as conditionally usable, many others fall into high-risk categories requiring substantial oversight. This study illustrates that the epistemic opacity of LLMs presents serious challenges for clinical deployment and reinforces the need for domain-specific, multi-criteria evaluation frameworks. In this context, uncertainty is as much a sociotechnical phenomenon as a computational one—emerging from the interaction between statistical properties, interface design, and professional expectations.

Conclusion

Collectively, the contributions to this special issue make a compelling case that uncertainty—when properly modeled, interpreted, and communicated—is not an impediment to clinical AI but its necessary foundation. These papers span a wide methodological spectrum, from causal reasoning and anomaly detection to explainability and human-centered design. Yet, they converge on a shared insight: that trust, safety, and utility in medical AI depend not only on predictive accuracy, but on the system's capacity to represent what it does not know, to signal when it may be wrong, and to support human users in interpreting its limits. As the integration of machine learning into clinical workflows continues to accelerate, it is imperative that uncertainty be treated not as a marginal consideration but as a central design principle—one that informs technical architectures, user interfaces, evaluation metrics, and institutional practices alike. In doing so, we move toward a more reflective, transparent, and ultimately responsible paradigm for AI in healthcare.

CRediT authorship contribution statement

Andrea Campagner: Conceptualization, Writing – original draft, Writing – review & editing. **Elia Mario Biganzoli:** Writing – review & editing. **Clara Balsano:** Writing – review & editing. **Cristina Cereda:** Writing – review & editing. **Federico Cabitza:** Conceptualization, Writing – original draft, Writing – review & editing.

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